



Demo Projects Overview

Description:

List of demo projects that you are building throughout the DevOps Bootcamp.

Purpose:

This should make it easier for you to keep track of building different Git repos and list them in your portfolio, e.g. for your job search.

Module 5 - Cloud & Infrastructure as Service Basics



Demo Project:

Create server and deploy application on DigitalOcean

Technologies used:

DigitalOcean, Linux, Java, Gradle

Project Description:

- Setup and configure a server on DigitalOcean
- Create and configure a new Linux user on the Droplet (Security best practice)
- Deploy and run a Java Gradle application on Droplet

Module 5: Cloud & Infrastructure as Service Basics

Module 6 - Artifact Repository Manager with Nexus



Demo Project:

Run Nexus on Droplet and Publish Artifact to Nexus

Technologies used:

Nexus, DigitalOcean, Linux, Java, Gradle, Maven

Project Description:

- Install and configure Nexus from scratch on a cloud server
- Create new User on Nexus with relevant permissions
- Java Gradle Project: Build Jar & Upload to Nexus
- Java Maven Project: Build Jar & Upload to Nexus

Module 6: Artifact Repository Manager with Nexus

Module 7 - Containers with Docker

Demo Project:

Use Docker for local development



Technologies used:

Docker, Node.js, MongoDB, MongoExpress

Project Description:

- Create Dockerfile for Nodejs application and build Docker image
- Run Nodejs application in Docker container and connect to MongoDB database container locally.
- Also run MongoExpress container as a UI of the MongoDB database.

Module 7: Containers with Docker

Demo Project:

Docker Compose – Run multiple Docker containers



Technologies used:

Docker, MongoDB, MongoExpress

Project Description:

- Write Docker Compose file to run MongoDB and MongoExpress containers

Module 7: Containers with Docker

Demo Project:

Dockerize Nodejs application



Technologies used:

Docker, Node.js,

Project Description:

- Write Dockerfile to build a Docker image for a Nodejs application

Module 7: Containers with Docker

Demo Project:

Persist data with Docker Volumes



Technologies used:

Docker, Node.js, MongoDB

Project Description:

- Persist data of a MongoDB container by attaching a Docker volume to it

Module 7: Containers with Docker

Demo Project:

Create Docker repository on Nexus and push to it



Technologies used:

Docker, Nexus, DigitalOcean, Linux

Project Description:

- Create Docker hosted repository on Nexus
- Create Docker repository role on Nexus
- Configure Nexus, DigitalOcean Droplet and Docker to be able to push to Docker repository
- Build and Push Docker image to Docker repository on Nexus

Module 7: Containers with Docker

Demo Project:

Deploy Docker application on a server with Docker Compose



Technologies used:

Docker, Amazon ECR, Node.js, MongoDB, MongoExpress

Project Description:

- Copy Docker-compose file to remote server
- Login to private Docker registry on remote server to fetch our app image
- Start our application container with MongoDB and MongoExpress services using docker compose

Module 7: AWS Services

Demo Project:

Deploy Nexus as Docker container



Technologies used:

Docker, Nexus, DigitalOcean, Linux

Project Description:

- Create and Configure Droplet
- Set up and run Nexus as a Docker container

Module 7: Containers with Docker

Module 8 - Build Automation & CI/CD with Jenkins



Demo Project:

Install Jenkins on DigitalOcean

Technologies used:

Jenkins, Docker, DigitalOcean, Linux

Project Description:

- Create an Ubuntu server on DigitalOcean
- Set up and run Jenkins as Docker container
- Initialize Jenkins

Module 8: Build Automation & CI/CD with Jenkins



Demo Project:

Create a CI Pipeline with Jenkinsfile (Freestyle, Pipeline, Multibranch Pipeline)

Technologies used:

Jenkins, Docker, Linux, Git, Java, Maven

Project Description:

CI Pipeline for a Java Maven application to build and push to the repository

- Install Build Tools (Maven, Node) in Jenkins
- Make Docker available on Jenkins server
- Create Jenkins credentials for a git repository
- Create different Jenkins job types (Freestyle, Pipeline, Multibranch pipeline) for the Java Maven project with Jenkinsfile to:
 - a. Connect to the application's git repository
 - b. Build Jar
 - c. Build Docker Image
 - d. Push to private DockerHub repository

Module 8: Build Automation & CI/CD with Jenkins



Demo Project:

Create a Jenkins Shared Library

Technologies used:

Jenkins, Groovy, Docker, Git, Java, Maven

Project Description:

Create a Jenkins Shared Library to extract common build logic:

- Create separate Git repository for Jenkins Shared Library project
- Create functions in the JSL to use in the Jenkins pipeline
- Integrate and use the JSL in Jenkins Pipeline (globally and for a specific project in Jenkinsfile)

Module 8: Build Automation & CI/CD with Jenkins



Demo Project:

Configure Webhook to trigger CI Pipeline automatically on every change

Technologies used:

Jenkins, GitLab, Git, Docker, Java, Maven

Project Description:

- Install GitLab Plugin in Jenkins
- Configure GitLab access token and connection to Jenkins in GitLab project settings
- Configure Jenkins to trigger the CI pipeline, whenever a change is pushed to GitLab

Module 8: Build Automation & CI/CD with Jenkins



Demo Project:

Dynamically Increment Application version in Jenkins Pipeline

Technologies used:

Jenkins, Docker, GitLab, Git, Java, Maven

Project Description:

- Configure CI step: Increment patch version
- Configure CI step: Build Java application and clean old artifacts
- Configure CI step: Build Image with dynamic Docker Image Tag
- Configure CI step: Push Image to private DockerHub repository
- Configure CI step: Commit version update of Jenkins back to Git repository
- Configure Jenkins pipeline to not trigger automatically on CI build commit to avoid commit loop

Module 8: Build Automation & CI/CD with Jenkins

Module 9 - AWS Services



Demo Project:

Deploy Web Application on EC2 Instance (manually)

Technologies used:

AWS, Docker, Linux

Project Description:

- Create and configure an EC2 Instance on AWS
- Install Docker on remote EC2 Instance
- Deploy Docker image from private Docker repository on EC2 Instance

Module 9: AWS Services



Demo Project:

CD - Deploy Application from Jenkins Pipeline to EC2 Instance (automatically with docker)

Technologies used:

AWS, Jenkins, Docker, Linux, Git, Java, Maven, Docker Hub

Project Description:

- Prepare AWS EC2 Instance for deployment (Install Docker)
- Create ssh key credentials for EC2 server on Jenkins
- Extend the previous CI pipeline with deploy step to ssh into the remote EC2 instance and deploy newly built image from Jenkins server
- Configure security group on EC2 Instance to allow access to our web application

Module 9: AWS Services



Demo Project:

CD - Deploy Application from Jenkins Pipeline on EC2 Instance (automatically with docker-compose)

Technologies used:

AWS, Jenkins, Docker, Linux, Git, Java, Maven, Docker Hub

Project Description:

- Install Docker Compose on AWS EC2 Instance
- Create docker-compose.yml file that deploys our web application image
- Configure Jenkins pipeline to deploy newly built image using Docker Compose on EC2 server
- Improvement: Extract multiple Linux commands that are executed on remote server into a separate shell script and execute the script from Jenkinsfile

Module 9: AWS Services



Demo Project:

Complete the CI/CD Pipeline (Docker-Compose, Dynamic versioning)

Technologies used:

AWS, Jenkins, Docker, Linux, Git, Java, Maven, Docker Hub

Project Description:

- CI step: Increment version
- CI step: Build artifact for Java Maven application
- CI step: Build and push Docker image to Docker Hub
- CD step: Deploy new application version with Docker Compose
- CD step: Commit the version update

Module 9: AWS Services



Demo Project:

Create repository on AWS and push to private Docker registry

Technologies used:

Docker, Amazon ECR

Project Description:

- Create private Docker registry on AWS (Amazon ECR)
- Tag and Push Docker image to this private repository

Module 9: AWS Services



Demo Project:

Interacting with AWS CLI

Technologies used:

AWS, Linux

Project Description:

- Install & configure AWS CLI to connect to our AWS account
- Create EC2 Instance using AWS CLI with all configurations like Security Group
- Create SSH key pair
- Create IAM resources like User, Group, Policy using the AWS CLI
- List and browse AWS resources using the AWS CLI

Module 9: AWS Services

Module 10 - Container Orchestration with Kubernetes



Demo Project:

Deploy MongoDB and Mongo Express into local K8s cluster

Technologies used:

Kubernetes, Docker, MongoDB, Mongo Express

Project Description:

- Setup local K8s cluster with Minikube
- Deploy MongoDB and MongoExpress with configuration and credentials extracted into ConfigMap and Secret

Module 10: Container Orchestration with Kubernetes



Demo Project:

Deploy Mosquitto message broker with ConfigMap and Secret Volume Types

Technologies used:

Kubernetes, Docker, Mosquitto

Project Description:

- Define configuration and passwords for Mosquitto message broker with ConfigMap and Secret Volume types

Module 10: Container Orchestration with Kubernetes



Demo Project:

Install a stateful service (MongoDB) on Kubernetes using Helm

Technologies used:

K8s, Helm, MongoDB, Mongo Express, Linode LKE, Linux

Project Description:

- Create a managed K8s cluster with Linode Kubernetes Engine
- Deploy replicated MongoDB service in LKE cluster using a Helm chart
- Configure data persistence for MongoDB with Linode's cloud storage
- Deploy UI client Mongo Express for MongoDB
- Deploy and configure nginx ingress to access the UI application from browser

Module 10: Container Orchestration with Kubernetes



Demo Project:

Deploy our web application in K8s cluster from private Docker registry

Technologies used:

Kubernetes, Helm, AWS ECR, Docker

Project Description:

- Create Secret for credentials for the private Docker registry
- Configure the Docker registry secret in application Deployment component
- Deploy web application image from our private Docker registry in K8s cluster

Module 10: Container Orchestration with Kubernetes



Demo Project:

Deploy Microservices application in Kubernetes with Production & Security Best Practices

Technologies used:

Kubernetes, Redis, Linux, Linode LKE

Project Description:

- Create K8s manifests for Deployments and Services for all microservices of an online shop application
- Deploy microservices to Linode's managed Kubernetes cluster

Module 10: Container Orchestration with Kubernetes

**Demo Project:**

Create Helm Chart for Microservices

Technologies used:

Kubernetes, Helm

Project Description:

- Create 1 shared Helm Chart for all microservices, to reuse common Deployment and Service configurations for the services

Module 10: Container Orchestration with Kubernetes

**Demo Project:**

Deploy Microservices with Helmfile

Technologies used:

Kubernetes, Helm, Helmfile

Project Description:

- Deploy Microservices with Helm
- Deploy Microservices with Helmfile

Module 10: Container Orchestration with Kubernetes

**Demo Project:**

Create AWS EKS cluster with a Node Group

Technologies used:

Kubernetes, AWS EKS

Project Description:

- Configure necessary IAM Roles
- Create VPC with Cloudformation Template for Worker Nodes
- Create EKS cluster (Control Plane Nodes)
- Create Node Group for Worker Nodes and attach to EKS cluster
- Configure Auto-Scaling of worker nodes
- Deploy a sample application to EKS cluster

Module 11: Kubernetes on AWS - EKS

**Demo Project:**

Create EKS cluster with Fargate profile

Technologies used:

Kubernetes, AWS EKS, AWS Fargate

Project Description:

- Create Fargate IAM Role
- Create Fargate Profile
- Deploy an example application to EKS cluster using Fargate profile

Module 11: Kubernetes on AWS - EKS

**Demo Project:**

Create EKS cluster with eksctl

Technologies used:

Kubernetes, AWS EKS, Eksctl, Linux

Project Description:

- Create EKS cluster using eksctl tool that reduces the manual effort of creating an EKS cluster

Module 11: Kubernetes on AWS - EKS

**Demo Project:**

CD - Deploy to EKS cluster from Jenkins Pipeline

Technologies used:

Kubernetes, Jenkins, AWS EKS, Docker, Linux

Project Description:

- Install kubectl and aws-iam-authenticator on a Jenkins server
- Create kubeconfig file to connect to EKS cluster and add it on Jenkins server
- Add AWS credentials on Jenkins for AWS account authentication
- Extend and adjust Jenkinsfile of the previous CI/CD pipeline to configure connection to EKS cluster

Module 11: Kubernetes on AWS - EKS

**Demo Project:**

CD - Deploy to LKE cluster from Jenkins Pipeline

Technologies used:

Kubernetes, Jenkins, Linode LKE, Docker, Linux

Project Description:

- Create K8s cluster on LKE
- Install kubectl as Jenkins Plugin
- Adjust Jenkinsfile to use Plugin and deploy to LKE cluster

Module 11: Kubernetes on AWS - EKS

**Demo Project:**

Complete CI/CD Pipeline with EKS and private DockerHub registry

Technologies used:

Kubernetes, Jenkins, AWS EKS, Docker Hub, Java, Maven, Linux, Docker, Git

Project Description:

- Write K8s manifest files for Deployment and Service configuration
- Integrate deploy step in the CI/CD pipeline to deploy newly built application image from DockerHub private registry to the EKS cluster
- So the complete CI/CD project we build has the following configuration:
 - a. CI step: Increment version
 - b. CI step: Build artifact for Java Maven application
 - c. CI step: Build and push Docker image to DockerHub
 - d. CD step: Deploy new application version to EKS cluster
 - e. CD step: Commit the version update

Module 11: Kubernetes on AWS - EKS

**Demo Project:**

Complete CI/CD Pipeline with EKS and AWS ECR

Technologies used:

Kubernetes, Jenkins, AWS EKS, AWS ECR, Java, Maven, Linux, Docker, Git

Project Description:

- Create private AWS ECR Docker repository
- Adjust Jenkinsfile to build and push Docker Image to AWS ECR
- Integrate deploying to K8s cluster in the CI/CD pipeline from AWS ECR private registry
- So the complete CI/CD project we build has the following configuration:
 - a. CI step: Increment version
 - b. CI step: Build artifact for Java Maven application
 - c. CI step: Build and push Docker image to AWS ECR
 - d. CD step: Deploy new application version to EKS cluster
 - e. CD step: Commit the version update

Module 11: Kubernetes on AWS - EKS

Module 12 - Infrastructure as Code with Terraform



Demo Project:

Automate AWS Infrastructure

Technologies used:

Terraform, AWS, Docker, Linux, Git

Project Description:

- Create TF project to automate provisioning AWS Infrastructure and its components, such as: VPC, Subnet, Route Table, Internet Gateway, EC2, Security Group
- Configure TF script to automate deploying Docker container to EC2 instance

Module 12: Infrastructure as Code with Terraform



Demo Project:

Modularize Project

Technologies used:

Terraform, AWS, Docker, Linux, Git

Project Description:

- Divide Terraform resources into reusable modules

Module 12: Infrastructure as Code with Terraform



Demo Project:

Terraform & AWSEKS

Technologies used:

Terraform, AWS EKS, Docker, Linux, Git

Project Description:

- Automate provisioning EKS cluster with Terraform

Module 12: Infrastructure as Code with Terraform



Demo Project:

Configure a Shared Remote State

Technologies used:

Terraform, AWS S3

Project Description:

- Configure Amazon S3 as remote storage for Terraform state

Module 12: Infrastructure as Code with Terraform



Demo Project:

Complete CI/CD with Terraform

Technologies used:

Terraform, Jenkins, Docker, AWS, Git, Java, Maven, Linux, Docker Hub

Project Description:

Integrate provisioning stage into complete CI/CD Pipeline to automate provisioning server instead of deploying to an existing server

- Create SSHKey Pair
- Install Terraform inside Jenkins container
- Add Terraform configuration to application's git repository
- Adjust Jenkinsfile to add "provision" step to the CI/CD pipeline that provisions EC2 instance
- So the complete CI/CD project we build has the following configuration:
 - a. CI step: Build artifact for Java Maven application
 - b. CI step: Build and push Docker image to Docker Hub
 - c. CD step: Automatically provision EC2 instance using TF
 - d. CD step: Deploy new application version on the provisioned EC2 instance with Docker Compose

Module 12: Infrastructure as Code with Terraform

Module 13 - Programming with Python

**Demo Project:**

Write Countdown Application

Technologies used:

Python, IntelliJ, Git

Project Description:

- Write an application that accepts a user input of a goal and a deadline (date). Print the remaining time until that deadline

Module 13: Programming with Python

**Demo Project:**

Automation with Python

Technologies used:

Python, IntelliJ, Git

Project Description:

- Write an application that reads a spreadsheet file and process and manipulate the spreadsheet

Module 13: Programming with Python

**Demo Project:**

API Request to GitLab

Technologies used:

Python, GitLab, IntelliJ, Git

Project Description:

- Write an application that talks to an API of an external application (GitLab) and lists all the public GitLab repositories for a specified user

Module 13: Programming with Python

Module 14 - Automation with Python



Demo Project:

Health Check: EC2 Status Checks

Technologies used:

Python, Boto3, AWS, Terraform

Project Description:

- Create EC2 Instances with Terraform
- Write a Python script that fetches statuses of EC2 Instances and prints to the console
- Extend the Python script to continuously check the status of EC2 Instances in a specific interval

Module 14: Automation with Python



Demo Project:

Automate configuring EC2 Server Instances

Technologies used:

Python, Boto3, AWS

Project Description:

- Write a Python script that automates adding environment tags to all EC2 Server instances

Module 14: Automation with Python



Demo Project:

Automate displaying EKS cluster information

Technologies used:

Python, Boto3, AWSEKS

Project Description:

- Write a Python script that fetches and displays EKS cluster status and information

Module 14: Automation with Python



Demo Project:

Data Backup & Restore

Technologies used:

Python, Boto3, AWS

Project Description:

- Write a Python script that automates creating backups for EC2 Volumes
- Write a Python script that cleans up old EC2 Volume snapshots
- Write a Python script that restores EC2 Volumes

Module 14: Automation with Python



Demo Project:

Website Monitoring and Recovery

Technologies used:

Python, Linode, Docker, Linux

Project Description:

- Create a server on a cloud platform
- Install Docker and run a Docker container on the remote server
- Write a Python script that monitors the website by accessing it and validating the HTTP response
- Write a Python script that sends an email notification when website is down
- Write a Python script that automatically restarts the application & server when the application is down

Module 14: Automation with Python

Module 15 - Configuration Management with Ansible



Demo Project:

Automate Node.js application Deployment

Technologies used:

Ansible, Node.js, DigitalOcean, Linux

Project Description:

- Create Server on DigitalOcean
- Write Ansible Playbook that installs necessary technologies, creates Linux user for an application and deploys a NodeJS application with that user

Module 15: Configuration Management with Ansible



Demo Project:

Automate Nexus Deployment

Technologies used:

Ansible, Nexus, DigitalOcean, Java, Linux

Project Description:

- Create Server on DigitalOcean
- Write Ansible Playbook that creates Linux user for Nexus, configure server, installs and deploys Nexus and verifies that it is running successfully

Module 15: Configuration Management with Ansible



Demo Project:

Ansible & Docker

Technologies used:

Ansible, AWS, Docker, Terraform, Linux

Project Description:

- Create AWS EC2 Instance with Terraform
- Write Ansible Playbook that installs necessary technologies like Docker and Docker Compose, copies docker-compose file to the server and starts the Docker containers configured inside the docker-compose file

Module 15: Configuration Management with Ansible



Demo Project:

Ansible Integration in Terraform

Technologies used:

Ansible, Terraform, AWS, Docker, Linux

Project Description:

- Create Ansible Playbook for Terraform integration
- Adjust Terraform configuration to execute Ansible Playbook automatically, so once Terraform provisions a server, it executes an Ansible playbook that configures the server

Module 15: Configuration Management with Ansible



Demo Project:

Configure Dynamic Inventory

Technologies used:

Ansible, Terraform, AWS

Project Description:

- Create EC2 Instance with Terraform
- Write Ansible AWS EC2 Plugin to dynamically sets inventory of EC2 servers that Ansible should manage, instead of hard-coding server addresses in Ansible inventory file

Module 15: Configuration Management with Ansible



Demo Project:

Automate Kubernetes Deployment

Technologies used:

Ansible, Terraform, Kubernetes, AWS EKS, Python, Linux

Project Description:

- Create EKS cluster with Terraform
- Write Ansible Play to deploy application in a new K8s namespace

Module 15: Configuration Management with Ansible

Module 15 - Configuration Management with Ansible



Demo Project:

Ansible Integration in Jenkins

Technologies used:

Ansible, Jenkins, DigitalOcean, AWS, Boto3, Docker, Java, Maven, Linux, Git

Project Description:

- Create and configure a dedicated server for Jenkins
- Create and configure a dedicated server for Ansible Control Node
- Write Ansible Playbook, which configures 2 EC2 Instances
- Add ssh key file credentials in Jenkins for Ansible Control Node server and Ansible Managed Node servers
- Configure Jenkins to execute the Ansible Playbook on remote Ansible Control Node server as part of the CI/CD pipeline
- So the Jenkinsfile configuration will do the following:
 - a. Connect to the remote Ansible Control Node server
 - b. Copy Ansible playbook and configuration files to the remote Ansible Control Node server
 - c. Copy the ssh keys for the Ansible Managed Node servers to the Ansible Control Node server
 - d. Install Ansible, Python3 and Boto3 on the Ansible Control Node server
 - e. With everything installed and copied to the remote Ansible Control Node server, execute the playbook remotely on that Control Node that will configure the 2 EC2 Managed Nodes

Module 15: Configuration Management with Ansible



Demo Project:

Structure Playbooks with Ansible Roles

Technologies used:

Ansible, Docker, AWS, Linux

Project Description:

- Break up large Ansible Playbooks into smaller manageable files using Ansible Roles

Module 15: Configuration Management with Ansible

Module 16 - Monitoring with Prometheus



Demo Project:

Install Prometheus Stack in Kubernetes

Technologies used:

Prometheus, Kubernetes, Helm, AWS EKS, eksctl, Grafana, Linux

Project Description:

- Setup EKS cluster using eksctl
- Deploy Prometheus, Alert Manager and Grafana in cluster as part of the Prometheus Operator using Helm chart

Module 16: Monitoring with Prometheus



Demo Project:

Configure Alerting for our Application

Technologies used:

Prometheus, Kubernetes, Linux

Project Description:

Configure our Monitoring Stack to notify us whenever CPU usage > 50% or Pod cannot start

- Configure Alert Rules in Prometheus Server
- Configure Alertmanager with Email Receiver

Module 16: Monitoring with Prometheus



Demo Project:

Configure Monitoring for a Third-Party Application

Technologies used:

Prometheus, Kubernetes, Redis, Helm, Grafana

Project Description:

Monitor Redis by using Prometheus Exporter

- Deploy Redis service in our cluster
- Deploy Redis exporter using Helm Chart
- Configure Alert Rules (when Redis is down or has too many connections)
- Import Grafana Dashboard for Redis to visualize monitoring data in Grafana

Module 16: Monitoring with Prometheus



Demo Project:

Configure Monitoring for Own Application

Technologies used:

Prometheus, Kubernetes, Node.js, Grafana, Docker, Docker Hub

Project Description:

- Configure our NodeJS application to collect & expose Metrics with Prometheus Client Library
- Deploy the NodeJS application, which has a metrics endpoint configured, into Kubernetes cluster
- Configure Prometheus to scrape this exposed metrics and visualize it in Grafana Dashboard

Module 16: Monitoring with Prometheus